LD10 Final Project Summary

Boosting project using Bank Marketing and House Prices datasets

XGBoost	0.8499	$24,772.56	$25,495.43
Best classification model	XGBoost F1-score	Best regression RMSE	Final ensemble RMSE

1. Project objective

The objective of Learning Day 10 was to learn and apply boosting methods for tabular business data. The project covers XGBoost, CatBoost, LightGBM, Optuna tuning and final model averaging. The work combines theory, practical modelling, model comparison and business interpretation.

2. Datasets used

Dataset	Problem type	Rows / columns	Business meaning
Bank Marketing	Classification	11,162 rows / 17 columns	Predict whether a customer will subscribe to a term deposit.
House Prices	Regression	1,460 rows / 81 columns	Predict property sale price from housing attributes.

3. Methods covered

- Bagging vs Boosting and Random Forest recap
- AdaBoost and Gradient Boosting foundation
- Forward stagewise additive modelling and pseudo-residuals
- XGBoost classification and regression
- CatBoost regression without manual encoding
- LightGBM tuning with Optuna
- Averaging ensemble using XGBoost, CatBoost and LightGBM

4. Key classification results - Bank Marketing

Model	Accuracy	F1-score	Interpretation
XGBoost Classifier	0.8542	0.8499	Best classification model in this run.
Random Forest Classifier	0.8470	0.8441	Strong baseline, but slightly weaker than XGBoost.

Classification conclusion

XGBoost achieved an F1-score of 0.8499, beating Random Forest by +0.0058. The improvement is small but meaningful because the model is used for customer conversion prediction, where balanced performance matters more than accuracy alone.

5. Feature importance result

Rank	Feature	Importance	Business meaning
1	duration	0.1820	Call duration was the strongest signal for deposit prediction.
2	contact	0.1469	The contact method strongly influenced conversion likelihood.
3	housing	0.1364	Housing loan status helped separate customer behaviour.

6. Key regression results - House Prices

Rank	Model	RMSE	Professional interpretation
1	XGBoost Baseline	$24,772.56	Best validation RMSE in this run. The simple, controlled XGBoost setup generalised best.
2	Averaging Ensemble	$25,495.43	Very close to tuned XGBoost and more stable than several single models.
3	XGBoost Tuned	$25,505.00	GridSearchCV selected a compact tree setup; validation RMSE was slightly weaker than baseline.
4	CatBoost	$26,664.89	Useful because it handles categorical columns directly, but RMSE was weaker here.
5	LightGBM Tuned	$26,858.08	Optuna improved LightGBM over baseline, but it did not beat XGBoost.
6	LightGBM Baseline	$28,990.67	Weakest regression model before tuning.

Regression conclusion

The best regression model was XGBoost Baseline with RMSE $24,772.56. The averaging ensemble was second with RMSE $25,495.43. This shows that model averaging improved stability but did not beat the best individual XGBoost model on this validation split.

7. Final recommendation

- Use XGBoost as the main model when strong tabular performance and feature importance are needed.
- Use CatBoost when the data has many categorical columns and preprocessing time must be reduced.

- Use LightGBM with Optuna when fast tuning is needed on larger datasets.
- Use averaging ensembles only when the small accuracy gain or stability improvement justifies extra model complexity.